

Clinic Appointment Booking System — Data Dictionary (Final)

Normalized to 3NF. Includes table purposes, NULL constraints, keys, master/lookup tables, and relationship cardinality.

ROLES				2 fields
Master list of user roles in the system.				
Field	Type	Null	Key	Notes
role_id	INT	NOT NULL	PK	1 = Patient, 2 = Doctor, 3 = Clinic, 4 = Clinic Staff, 5 = Admin
role_name	VARCHAR(20)	NOT NULL		e.g. Patient, Doctor, Clinic, Staff, Admin

USERS				9 fields
Core login/account table shared by every person in the system (patient, doctor, clinic owner, staff, admin).				
Field	Type	Null	Key	Notes
user_id	INT	NOT NULL	PK	
full_name	VARCHAR(100)	NOT NULL		
email	VARCHAR(100)	NOT NULL	UNIQUE	
phone	INT	NOT NULL	UNIQUE	kept as-is per guide's review
password_hash	VARCHAR(255)	NOT NULL		
address	TEXT	NULL		personal address; optional
role_id	INT	NOT NULL	FK → ROLES	replaces free-text role column
is_verified	BOOLEAN	NOT NULL		default FALSE; quick-lookup flag, full history in VERIFICATION_LOG
account_status	VARCHAR(20)	NOT NULL		default 'Active'; Active / Inactive / Suspended

STATE_MASTER				2 fields
Master list of states/provinces, used to support location-based search and filtering.				
Field	Type	Null	Key	Notes
state_id	INT	NOT NULL	PK	
state_name	VARCHAR(100)	NOT NULL		

CITY_MASTER				3 fields
Master list of cities, linked to their state, so clinics/doctors can be searched or filtered by location.				
Field	Type	Null	Key	Notes
city_id	INT	NOT NULL	PK	

CITY_MASTER				3 fields
state_id	INT	NOT NULL	FK → STATE_MASTER	
city_name	VARCHAR(100)	NOT NULL		

CLINICS				9 fields
<i>One row per registered clinic/hospital branch.</i>				
Field	Type	Null	Key	Notes
clinic_id	INT	NOT NULL	PK	
user_id	INT	NOT NULL	FK + UNIQUE	
clinic_name	VARCHAR(100)	NOT NULL		
address	TEXT	NOT NULL		street-level address only; city/state now separated out
city_id	INT	NOT NULL	FK → CITY_MASTER	enables search/filter by city
contact_number	INT	NOT NULL		kept as-is per guide's review
latitude	DECIMAL(10,7)	NULL		
longitude	DECIMAL(10,7)	NULL		
created_at / updated_at	DATETIME	NOT NULL		audit timestamps

opening_time / closing_time removed from CLINICS — moved to CLINIC_HOURS to support per-day hours.

CLINIC_HOURS				5 fields
<i>Per-day operating hours for each clinic.</i>				
Field	Type	Null	Key	Notes
hours_id	INT	NOT NULL	PK	
clinic_id	INT	NOT NULL	FK	
day_id	INT	NOT NULL	FK → DAY_MASTER	
open_time	TIME	NULL		null if closed that day
close_time	TIME	NULL		null if closed that day

DAY_MASTER				2 fields
<i>Master list of weekdays, shared by CLINIC_HOURS and DOCTOR_AVAILABILITY to avoid duplicate/inconsistent day-name text in two places.</i>				
Field	Type	Null	Key	Notes
day_id	INT	NOT NULL	PK	1 = Sunday ... 7 = Saturday

DAY_MASTER				2 fields
day_name	VARCHAR(10)	NOT NULL		

DOCTORS				8 fields
One row per doctor profile.				
Field	Type	Null	Key	Notes
doctor_id	INT	NOT NULL	PK	
user_id	INT	NOT NULL	FK + UNIQUE	
doctor_name	VARCHAR(100)	NOT NULL		
experience	INT	NULL		years
new_patient_fee	DECIMAL(10,2)	NOT NULL		replaces single consultation_fee
old_patient_fee	DECIMAL(10,2)	NOT NULL		follow-up fee
about	TEXT	NULL		
created_at / updated_at	DATETIME	NOT NULL		audit timestamps

specialization, qualification and clinic_id removed from DOCTORS — each is a many-to-many relationship via the junction tables below.

SPECIALIZATION_MASTER				2 fields
Controlled list of medical specializations.				
Field	Type	Null	Key	Notes
specialization_id	INT	NOT NULL	PK	
specialization_name	VARCHAR(100)	NOT NULL		e.g. Cardiology, Pediatrics, Dermatology

DOCTOR_SPECIALIZATION				3 fields
Junction table — one doctor can have many specializations, one specialization can belong to many doctors.				
Field	Type	Null	Key	Notes
id	INT	NOT NULL	PK	
doctor_id	INT	NOT NULL	FK → DOCTORS	
specialization_id	INT	NOT NULL	FK → SPECIALIZATION_MASTER	

QUALIFICATION_MASTER				2 fields
Controlled list of academic/medical qualifications (degrees), kept separate from specializations.				
Field	Type	Null	Key	Notes
qualification_id	INT	NOT	PK	

QUALIFICATION_MASTER				2 fields
		NULL		
qualification_name	VARCHAR(100)	NOT NULL		e.g. MBBS, MD, MS — a degree, not a specialization

DOCTOR_QUALIFICATION				3 fields
<i>Junction table — one doctor can hold many qualifications.</i>				
Field	Type	Null	Key	Notes
id	INT	NOT NULL	PK	
doctor_id	INT	NOT NULL	FK → DOCTORS	
qualification_id	INT	NOT NULL	FK → QUALIFICATION_MASTER	

DOCTOR_CLINIC				3 fields
<i>Junction table — supports a doctor practicing at more than one clinic.</i>				
Field	Type	Null	Key	Notes
id	INT	NOT NULL	PK	
doctor_id	INT	NOT NULL	FK → DOCTORS	
clinic_id	INT	NOT NULL	FK → CLINICS	

DOCTOR_AVAILABILITY				6 fields
<i>Weekly time slots when a doctor is available at a given clinic.</i>				
Field	Type	Null	Key	Notes
availability_id	INT	NOT NULL	PK	
doctor_id	INT	NOT NULL	FK	
clinic_id	INT	NOT NULL	FK	which clinic this slot applies to
day_id	INT	NOT NULL	FK → DAY_MASTER	
start_time	TIME	NOT NULL		
end_time	TIME	NOT NULL		

PATIENTS				7 fields
<i>Patient-specific profile data, linked 1:1 to a USERS record.</i>				
Field	Type	Null	Key	Notes

PATIENTS				7 fields
patient_id	INT	NOT NULL	PK	
user_id	INT	NOT NULL	FK + UNIQUE	1-to-1 link to USERS (login/shared data lives there)
date_of_birth	DATE	NOT NULL		
gender	VARCHAR(10)	NULL		
blood_group	VARCHAR(5)	NULL		
emergency_contact	INT	NULL		kept as-is per guide's review
created_at / updated_at	DATETIME	NOT NULL		audit timestamps

APPOINTMENTS				10 fields
<i>One row per booked appointment between a patient and a doctor at a clinic.</i>				
Field	Type	Null	Key	Notes
appointment_id	INT	NOT NULL	PK	
patient_id	INT	NOT NULL	FK	
doctor_id	INT	NOT NULL	FK	
clinic_id	INT	NOT NULL	FK	which clinic the visit is at
appointment_date	DATE	NOT NULL		
appointment_time	TIME	NOT NULL		
symptoms	TEXT	NULL		free text
status	VARCHAR(20)	NOT NULL		default 'Pending'; Pending · Confirmed · Rejected · Completed · Cancelled
patient_type	VARCHAR(10)	NOT NULL		New / Old (follow-up) — determines which fee applies
fee_charged	DECIMAL(10,2)	NOT NULL		snapshot of the fee at time of booking, so later rate changes don't alter past records

PAYMENTS				6 fields
<i>Payment record for a completed or pending appointment.</i>				
Field	Type	Null	Key	Notes
payment_id	INT	NOT NULL	PK	
appointment_id	INT	NOT NULL	FK + UNIQUE	
amount	DECIMAL(10,2)	NOT NULL		
payment_method	VARCHAR(20)	NOT NULL		Cash / UPI / Card, etc.

PAYMENTS				6 fields
payment_status	VARCHAR(20)	NOT NULL		default 'Pending'; Pending / Paid / Refunded
paid_at	DATETIME	NULL		null until paid

PATIENT_RECORDS				6 fields
<i>Diagnosis/medical record generated from a completed appointment.</i>				
Field	Type	Null	Key	Notes
record_id	INT	NOT NULL	PK	
patient_id	INT	NOT NULL	FK	
doctor_id	INT	NOT NULL	FK	
appointment_id	INT	NOT NULL	FK + UNIQUE	
diagnosis	TEXT	NULL		
notes	TEXT	NULL		

PRESCRIPTIONS				5 fields
<i>Prescription file uploaded for a given appointment.</i>				
Field	Type	Null	Key	Notes
prescription_id	INT	NOT NULL	PK	
appointment_id	INT	NOT NULL	FK + UNIQUE	
uploaded_by	INT	NOT NULL	FK → USERS	
file_path	VARCHAR(255)	NOT NULL		
created_at	DATETIME	NOT NULL		

REPORTS				6 fields
<i>Lab/diagnostic report file uploaded for a given appointment.</i>				
Field	Type	Null	Key	Notes
report_id	INT	NOT NULL	PK	
appointment_id	INT	NOT NULL	FK	
uploaded_by	INT	NOT NULL	FK → USERS	clinic staff (receptionist)
report_name	VARCHAR(100)	NOT NULL		
file_path	VARCHAR(255)	NOT NULL		

REPORTS				6 fields
created_at	DATETIME	NOT NULL		

REVIEWS				7 fields
<i>Patient review/rating left for a doctor after a completed appointment.</i>				
Field	Type	Null	Key	Notes
review_id	INT	NOT NULL	PK	
patient_id	INT	NOT NULL	FK	
doctor_id	INT	NOT NULL	FK	
appointment_id	INT	NOT NULL	FK + UNIQUE	
rating	DECIMAL(2, 1)	NOT NULL		
review_text	TEXT	NULL		
created_at	DATETIME	NOT NULL		

CLINIC_STAFF				3 fields
<i>Receptionist/staff accounts linked to a clinic.</i>				
Field	Type	Null	Key	Notes
staff_id	INT	NOT NULL	PK	
user_id	INT	NOT NULL	FK + UNIQUE	
clinic_id	INT	NOT NULL	FK	

ADMIN				3 fields
<i>System administrator accounts.</i>				
Field	Type	Null	Key	Notes
admin_id	INT	NOT NULL	PK	
user_id	INT	NOT NULL	FK + UNIQUE	
access_level	VARCHAR(20)	NOT NULL		e.g. Super Admin / Moderator

VERIFICATION_LOG				6 fields
<i>Audit trail of admin approval/rejection actions on doctor, clinic, and staff accounts.</i>				
Field	Type	Null	Key	Notes
verification_id	INT	NOT	PK	

VERIFICATION_LOG				6 fields
		NULL		
user_id	INT	NOT NULL	FK → USERS	the doctor / clinic / staff being verified
admin_id	INT	NOT NULL	FK → ADMIN	who approved/rejected
status	VARCHAR(20)	NOT NULL		Pending / Approved / Rejected
remarks	TEXT	NULL		reason, e.g. rejection notes
verified_at	DATETIME	NOT NULL		

Relationship / Cardinality Summary

Parent Table	Child Table	Cardinality	Relationship
ROLES	USERS	1 : M	one role applies to many users
STATE_MASTER	CITY_MASTER	1 : M	one state has many cities
CITY_MASTER	CLINICS	1 : M	one city has many clinics
USERS	PATIENTS / DOCTORS / CLINICS / CLINIC_STAFF / ADMIN	1 : 1	subtype profile per role
DOCTORS	DOCTOR_SPECIALIZATION	1 : M	M:N with SPECIALIZATION_MASTER via this junction
DOCTORS	DOCTOR_QUALIFICATION	1 : M	M:N with QUALIFICATION_MASTER via this junction
DOCTORS	DOCTOR_CLINIC	1 : M	M:N with CLINICS via this junction
DOCTORS	DOCTOR_AVAILABILITY	1 : M	one doctor has many weekly slots
CLINICS	CLINIC_HOURS	1 : M	one clinic has hours for each open day
DAY_MASTER	CLINIC_HOURS / DOCTOR_AVAILABILITY	1 : M	shared weekday reference
PATIENTS	APPOINTMENTS	1 : M	one patient can book many appointments
DOCTORS	APPOINTMENTS	1 : M	one doctor has many appointments
CLINICS	APPOINTMENTS	1 : M	one clinic hosts many appointments
APPOINTMENTS	PAYMENTS	1 : 1	one payment per appointment
APPOINTMENTS	PATIENT_RECORDS	1 : 1	one record per completed appointment
APPOINTMENTS	PRESCRIPTIONS	1 : 1	one prescription per appointment
APPOINTMENTS	REPORTS	1 : M	an appointment can have multiple report files
APPOINTMENTS	REVIEWS	1 : 1	one review per appointment
ADMIN	VERIFICATION_LOG	1 : M	one admin can process many verifications
USERS	VERIFICATION_LOG	1 : M	one user can be reviewed more than once (resubmission)

PK = Primary Key **FK = Foreign Key** **UNIQUE = one-to-one link** **Red text = fields flagged by guide, intentionally kept as INT**